

Handiko Gesang Anugrah Sejati

RETAIL ALGORITHMIC ANALYST & TRADER

Portfolio: handiko.github.io/algo/

Source Code: github.com/handiko

Focus: Quantitative Engineering

EXECUTIVE PROFILE

Quantitative-focused Algorithmic Trader and Systems Developer with a sophisticated understanding of market microstructure across Cryptocurrencies, Foreign Exchange (Forex), and Equities. Specialized in designing, building, and benchmarking high-fidelity automated execution systems and Expert Advisors (EAs) within the MQL5 ecosystem. Broad domain expertise emphasizes structural order book dynamics, stochastic liquidity modeling, stringent multi-tier risk mitigation frameworks, and algorithmic market making.

Proven capability in translating empirical statistical anomalies and probabilistic frameworks (e.g., Markov-Chain models, Monte Carlo optimizations, Machine Learning backtesting) into production-ready software solutions. Transitioning core technical capabilities in market mechanics into institutional Crypto Business Operations and Liquidity Management to structurally optimize exchange micro-liquidity, control toxic order flow, and maximize capital execution efficiency.

CORE TECHNICAL COMPETENCIES

EXECUTION SYSTEMS	QUANTITATIVE ANALYTICS	RISK & PORTFOLIO MANAGEMENT
MQL5 Expert Advisors	Machine Learning (XGBoost)	Monte Carlo Simulations
MetaTrader 5 Architecture	Order Book Dynamics	Optimal Frontier Modeling
Algorithmic Trade Automation	Market Microstructure	P&L Optimization
MQL5 Backtesting Suites	Liquidity Aggregation	Value-at-Risk (VaR)
Low-Latency Execution	Markov-Chain Analysis	Inventory Risk Management
Pine Script v5	Stochastic Modeling	Drawdown Mitigation

IN-DEPTH ENGINEERING PORTFOLIO

Category I: Machine Learning & Algorithmic Backtesting

Machine Learning Stock Backtesting Framework

[View Project →](#)

A comprehensive framework testing the efficacy of machine learning models in outperforming buy-and-hold strategies for Indonesian equity markets. Leverages XGBoost to classify market regimes and generate high-probability trade signals, incorporating rigorous backtesting pipelines to assess predictive performance vs. passive benchmarks.

ML & QUANTITATIVE FINANCE

Category II: Market Making, Liquidity Architecture & Exchange Simulation

Avellaneda-Stoikov Market Maker Simulator

[View Project →](#)

An interactive, high-fidelity deployment of the seminal Avellaneda-Stoikov (2008) framework modeling stochastic control problems for high-frequency liquidity providers. The system solves the continuous-time Hamilton-Jacobi-Bellman equation to dynamically compute inventory-adjusted optimal bid-ask reservation spreads.

STOCHASTIC CONTROL & SIMULATION

USDT/IDR Monitoring Terminal & Liquidity Dashboard

[View Project →](#)

A low-latency monitoring infrastructure designed as a "Terminal-as-a-Service" UI for the USDT/IDR stablecoin pair across regional Indonesian digital asset venues. Focuses heavily on deep-tier order book aggregation, microsecond order book tracking, and slip-page simulation.

LIQUIDITY INFRASTRUCTURE

Treasury Monitoring Terminal: Smart Rebalancing System

[View Project →](#)

A specialized exchange operations simulation application built to validate multi-venue capital rebalancing policies. Simulates automated exchange treasury heuristics designed to optimize reserve capital matching, hot/cold wallet distribution ratios, and operational liquidity buffers.

EXCHANGE OPERATIONS

Category III: Quantitative Portfolio Optimization & Strategic Performance Analytics

Multi-Criteria Optimal Portfolio Allocation Engine

[View Project →](#)

A programmatic Python-based capital allocation framework executing multi-thousand iteration Monte Carlo simulations to extract mathematically optimal asset weights. Moving aggressively beyond simplistic Mean-Variance frameworks, the model tracks multi-criteria metrics including Sortino ratios, Maximum Drawdown, and tail-risk indicators.

PORTFOLIO OPTIMIZATION

Quantitative Portfolio Performance Simulation and Analysis

[View Project →](#)

A comprehensive historical backtesting and statistical evaluation engine designed to rigorously analyze custom, data-driven optimal portfolios against major benchmark equity indices. Tracks and charts structural tracking errors, rolling annualized beta profiles, and idiosyncratic alpha generation.

PERFORMANCE BACKTESTING